# Homework 3

**Due: 5th February, 2026**

**Exercise 3.1** (Hogg, Tanis and Zimmerman, Exercises 6.1.7 and 6.2.10)**.** Ledolter and Hogg (see References in Hogg, Tanis and Zimmerman) report 64 observations that are a sample of daily weekday after-noon (3 to 7 p.m.) lead concentrations (in micrograms per cubic meter, $\mu g/m^3$). The following data were recorded at an air-monitoring station near the San Diego Freeway in Los Angeles during the fall of 1976:

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 6.7 | 5.4 | 5.2 | 6.0 | 8.7 | 6.0 | 6.4 | 8.3 | 5.3 | 5.9 | 7.6 |
| 5.0 | 6.9 | 6.8 | 4.9 | 6.3 | 5.0 | 6.0 | 7.2 | 8.0 | 8.1 | 7.2 |
| 10.9 | 9.2 | 8.6 | 6.2 | 6.1 | 6.5 | 7.8 | 6.2 | 8.5 | 6.4 | 8.1 |
| 2.1 | 6.1 | 6.5 | 7.9 | 14.1 | 9.5 | 10.6 | 8.4 | 8.3 | 5.9 | 6.0 |
| 6.4 | 3.9 | 9.9 | 7.6 | 6.8 | 8.6 | 8.5 | 11.2 | 7.0 | 7.1 | 6.0 |
| 9.0 | 10.1 | 8.0 | 6.8 | 7.3 | 9.7 | 9.3 | 3.2 | 6.4 | | |

1. Build a histogram for the above data set. Is the distribution symmetric?
2. Calculate the sample mean $\overline{x}$ and sample standard deviation $\sigma$.
3. Locate $\overline{x}$ and $\overline{x} \pm \sigma$ on your histogram. How many observations lie within one standard deviation of the mean?

During the fall of 1977, the weekday afternoon lead concentrations (in $\mu g/m^3$) at the measurement station near the San Diego Freeway in Los Angeles were as follows:

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 9.5 | 10.7 | 8.3 | 9.8 | 9.1 | 9.4 | 9.6 | 11.9 | 9.5 | 12.6 | 10.5 |
| 8.9 | 11.4 | 12.0 | 12.4 | 9.9 | 10.9 | 12.3 | 11.0 | 9.2 | 9.3 | 9.3 |
| 10.5 | 9.4 | 9.4 | 8.2 | 10.4 | 9.3 | 8.7 | 9.8 | 9.1 | 2.9 | 9.8 |
| 5.7 | 8.2 | 8.1 | 8.8 | 9.7 | 8.1 | 8.8 | 10.3 | 8.6 | 10.2 | 9.4 |
| 14.8 | 9.9 | 9.3 | 8.2 | 9.9 | 11.6 | 8.7 | 5.0 | 9.9 | 6.3 | 6.5 |
| 10.2 | 8.8 | 8.0 | 8.7 | 8.9 | 6.8 | 6.6 | 7.3 | 16.7 | | |

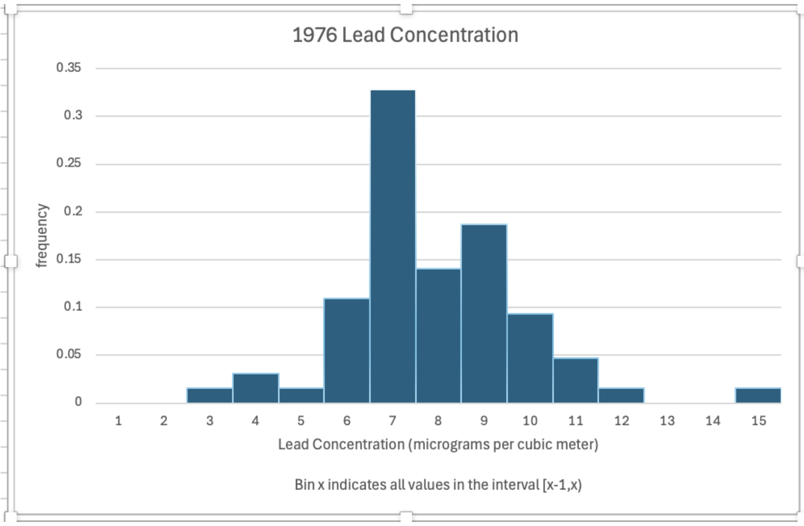
4. Build a histogram for the above data set. Is the distribution symmetric?
5. Calculate the sample mean $\overline{x}$ and sample standard deviation $\sigma$.
6. Locate $\overline{x}$ and $\overline{x} \pm \sigma$ on your histogram. How many observations lie within one standard deviation of the mean?
7. Construct box-and whisker plots of both the 1976 and 1977 data.
8. Use your numerical and graphical results to interpret what you see.

**Remark**: In the spring of 1977, a new traffic lane was added to the freeway. This lane reduced traffic congestion but increased traffic speed.

**See Next Page**

*Sol.*

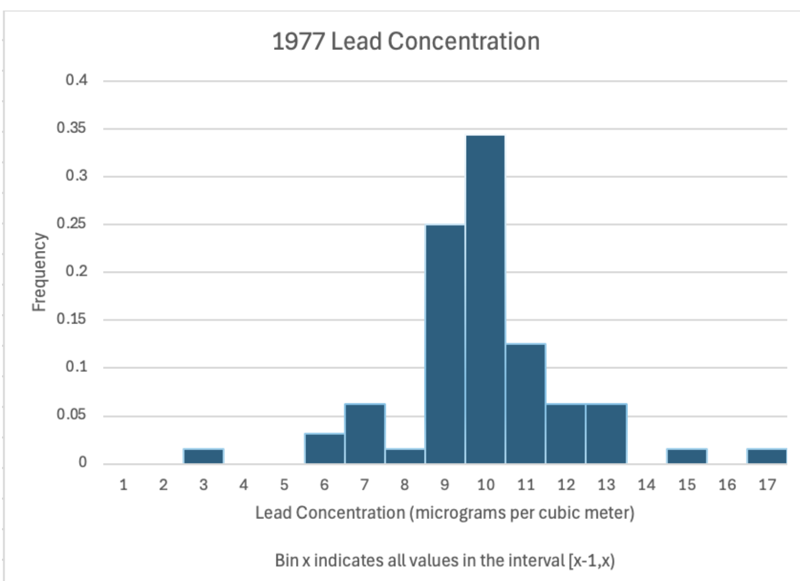
First, we have a histogram for the 1976 dataset:




I chose origin $\kappa = 8$ for my origin, but it doesn't really matter; $h = 1$ was my bin width, which is why the vertical axis is the literal frequency of occurrence in that bin (if you chose a different $h$, you would need to adjust the vertical axis so that the bin area was the frequency of occurrence in that bin). It is...not so symmetric. The peak in bin 7 really breaks the symmetry. The sample mean is $\boxed{\overline{x} = 7.275}$ and the sample standard deviation is $\boxed{\sigma \approx 1.97}$.

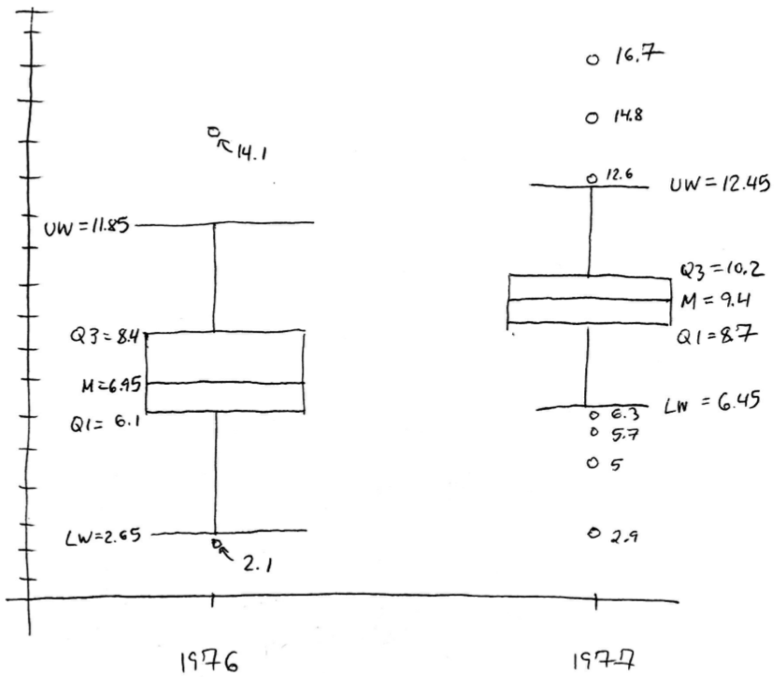
$\overline{x}$ lives in bin 8; $\overline{x} + \sigma$ lives in bin 10 and $\overline{x} - \sigma$ lives in bin 6. $\boxed{48}$ observations live within one standard deviation of the mean.

Next, we have a histogram for the 1977 dataset:




I chose origin $\kappa = 10$ for my origin, but it doesn't really matter; $h = 1$ was my bin width, which is why the vertical axis is the literal frequency of occurrence in that bin (if you chose a different $h$, you would need to adjust the vertical axis so that the bin area was the frequency of occurrence in that bin). It is...also not so symmetric. The sample mean is $\boxed{\overline{x} = 9.42}$ and the sample standard deviation is $\boxed{\sigma \approx 2.08}$.

$\overline{x}$ lives in bin 10, $\overline{x} + \sigma$ lives in bin 12 and $\overline{x} - \sigma$ lives in bin 8. $\boxed{50}$ observations live within one standard deviation of the mean, which is again the bulk of them. Below, we have box and whisker plots of both the 1976 and 1977 data:




(Yes I did them by hand ahaha - excel is a bit wonky about adjusting so it was easier to just draw it by hand). If your software package computes box and whisker plots in a slightly different way than the textbook, that's okay - I'm just looking for a sense of what a plot generally looks like, and the kinds of interpretation you can make from them.

It is clear that the dataset has shifted upwards: the interquartile ranges do not even overlap. The data is also more compressed (smaller IQR), which indicates more consistently high readings.

**Exercise 3.2** (Hogg, Tanis and Zimmerman, Exercise 5.4.22)**.** Let $X_1$ and $X_2$ be two independent random variables. Let $X_1 \sim \chi^2(r_1)$ and $Y = X_1 + X_2 \sim \chi^2(r)$, where $r > r_1$.

1. Find the moment generating function of $X_2$.
2. What is the distribution of $X_2$?

*Sol.* For the first item, observe that

$$M_Y(t) = \mathbb{E}\left[e^{tY}\right] = \mathbb{E}[\left[e^{t(X_1+X_2)}\right] = \mathbb{E}\left[e^{tX_1}e^{tX_2}\right] = \mathbb{E}\left[e^{tX_1}\right]\mathbb{E}\left[e^{tX_2}\right] = M_{X_1}(t)M_{X_2}(t)$$

using independence. Now, since $X_1 \sim \chi^2(r_1)$ and $Y \sim \chi^2(r)$, we find

$$(1-2t)^{-r/2} = (1-2t)^{-r_1/2}M_{X_2}(t) \implies \boxed{M_{X_2}(t) = (1-2t)^{-(r-r_1)/2}}$$

for any $t < \frac{1}{2}$. This is the moment generating function of a $\chi^2(r-r_1)$ random variable, so $\boxed{X_2 \sim \chi^2(r-r_1)}$.

**Exercise 3.3.** Recall that the *entropy* of a continuous probability distribution with probability density function $f(x)$ is defined by

$$H(f) := -\int_{-\infty}^{\infty} f(x)\ln f(x)\ dx. \tag{1}$$

1. Let $g$ be the probability density function of a $\mathcal{N}(0,1)$ random variable. Find $H(g)$.
2. Let $f$ be the probability density function of any random variable (not necessarily Gaussian) with mean $\mu = 0$ and variance $\sigma^2 = 1$. Find
$$-\int_{-\infty}^{\infty} f(x)\ln g(x)\ dx,$$
where $g$ is as in the previous part.

The **Kullback-Leibler (KL) divergence** is a fundamental quantity in information theory and statistics used to measure the distance between two probability density functions $f$ and $g$. It is given by

$$D_{KL}(f\|g) := \int_{-\infty}^{\infty} f(x)\ln\left(\frac{f(x)}{g(x)}\right)\ dx. \tag{2}$$

A fundamental result (Gibbs' inequality) says that for any two pdfs $f$ and $g$, $D_{KL}(f\|g) \geq 0$.

3. Using $D_{KL}(f\|g) \geq 0$ and your results from the first two parts of this exercise, show that for any probability density function $f(x)$ with mean zero and variance one,
$$H(f) \leq H(g)$$
where $g$ is the pdf of a $\mathcal{N}(0,1)$ random variable.
4. Look up the second law of thermodynamics, and in interpret the Central Limit Theorem in the context of this law and your above result.

*Sol.*

For the first item, recall that the density of a standard Gaussian is given by $g(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$. Hence,

$$H(g) = -\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}\ln\left(\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}\right)\ dx$$

$$
\begin{aligned}
&= -\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \ln\left(\frac{1}{\sqrt{2\pi}}\right)\ dx - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \ln\left(e^{-\frac{x^2}{2}}\right)\ dx \\
&= \ln(\sqrt{2\pi}) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\ dx + \frac{1}{2}\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} x^2\ dx \\
&= \ln(\sqrt{2\pi}) \int_{-\infty}^{\infty} g(x)\ dx + \frac{1}{2}\int_{-\infty}^{\infty} g(x) x^2\ dx
\end{aligned}
$$

where we have reinserted the definition of $g$. The reason for this is to specify that it doesn't actually matter what $g$ is now; the integral $\int_{-\infty}^{\infty} g(x)\ dx$ is just one because $g$ is a probability density function, and the integral $\int_{-\infty}^{\infty} g(x)x^2\ dx$ is the variance of the random variable whose density is $g$ ($\mathcal{N}(0,1)$), i.e. one. Hence,

$$H(g) = \boxed{\ln(\sqrt{2\pi}) + \frac{1}{2}}.$$

For part 2, the exact computation, under the assumption that $f$ is a density of a mean zero and variance one random variable, yields

$$
\begin{aligned}
-\int_{-\infty}^{\infty} f(x)\ln g(x)\ dx &= -\int_{-\infty}^{\infty} f(x)\ln\left(\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\right)\ dx \\
&= -\int_{-\infty}^{\infty} f(x)\ln\left(\frac{1}{\sqrt{2\pi}}\right)\ dx - \int_{-\infty}^{\infty} f(x)\ln\left(e^{-\frac{x^2}{2}}\right)\ dx \\
&= \ln(\sqrt{2\pi}) \int_{-\infty}^{\infty} f(x)\ dx + \frac{1}{2}\int_{-\infty}^{\infty} f(x)x^2\ dx \\
&= \ln(\sqrt{2\pi}) + \frac{1}{2}\mathbb{E}_f\left[X^2\right] = \ln(\sqrt{2\pi}) + \frac{1}{2}\mathrm{Var}_f(X) = \boxed{\ln(\sqrt{2\pi}) + \frac{1}{2}}.
\end{aligned}
$$

Now, for part 3, if we assume that the Kullback-Leibler divergence is nonnegative, then we can write (using properties of logarithms) that

$$
\begin{aligned}
0 \le D_{KL}(f\|g) := \int_{-\infty}^{\infty} f(x)\ln\left(\frac{f(x)}{g(x)}\right)\ dx = \int_{-\infty}^{\infty} f(x)\ln f(x)\ dx - \int_{-\infty}^{\infty} f(x)\ln g(x)\ dx \\
= -H(f) - \int_{-\infty}^{\infty} f(x)\ln g(x)\ dx.
\end{aligned}
$$

If we let $f$ be a generic density representing a mean-zero variance one random variable and $g$ be the density of a $\mathcal{N}(0,1)$ random variable, then

$$-\int_{-\infty}^{\infty} f(x)\ln g(x)\ dx = \ln(\sqrt{2\pi}) + \frac{1}{2} = H(g)$$

using the first two parts of this exercise, so

$$0 \le -H(f) - \int_{-\infty}^{\infty} f(x)\ln g(x)\ dx \implies H(f) \le -\int_{-\infty}^{\infty} f(x)\ln g(x)\ dx = \ln(\sqrt{2\pi}) + \frac{1}{2} = H(g),$$

as desired.

So, amongst all random variables with a given mean and variance, the entropy is maximized by a Gaussian of that mean and variance. Now, suppose $X_i$ are i.i.d. with mean $\mu$ and variance $\sigma^2$. Then, the CLT says that

$$\frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \to \mathcal{N}(0,1).$$

Now, $\sum_{i=1}^n X_i$ has mean $n\mu$ and variance $n\sigma^2$, so by the shift properties of mean and variance, $\frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}$ has mean zero and variance one for all $n$. The CLT then says that as $n$ increases, this quotient converges to the random variable that maximizes the entropy, which accords with the second law of thermodynamics prediction that the entropy of a process always increases.